Diverse Additive Models for Material Deterioration Regression : TabNAMs and Weighted Ensemble

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Abstract

Material deterioration prediction is critical for preventive maintenance activities, yet existing models struggle to balance interpretability and predictive performance. We propose a framework of **Diverse Additive Models** that extends classical GAMs by integrating multiple function families: spline-based statistical GAMs, tree-based Feature Additive Models (FAM), gradient-boosted Additive Models (GBAM), and TabNet-based Neural Additive Models (TabNAM). Using 45,058 bridge inspection records, we demonstrate that TabNAM achieves $R^2 = 0.215$ individually, while an optimized ensemble of GAM+TabNAM reaches $R^2 = 0.216$. This complementary combination of statistical models and deep learning preserves the interpretability of additive structures while capturing complex nonlinear relationships. Our framework provides actionable insights for infrastructure managers through partial dependence plots and feature importance analysis, enabling data-driven maintenance decision support.

Keywords: Diverse Additive Models, Neural Additive Models, Material Deterioration, Interpretable Machine Learning, Ensemble Weight Learning, TabNAMs, Bayesian Optimization.

1 Introduction

1.1 Background and Motivation

Material deterioration prediction is crucial across diverse engineering domains—from civil infrastructure (bridges, roads, pipelines) to manufacturing equipment and building components. Accurate deterioration forecasting enables preventive maintenance, optimizes resource allocation, and ensures structural safety. Traditional approaches rely on mechanistic models or expert judgment, leading to reactive rather than proactive maintenance strategies.

Diverse Additive Models [10, 24] offer a middle ground between interpretable linear models and black-box machine learning by decomposing predictions into smooth univariate functions:

$$g(\mathbb{E}[y]) = \beta_0 + \sum_{j=1}^{p} f_j(x_j) \tag{1}$$

where g is a link function and f_j are smooth functions (typically splines). This additive structure enables visualization of each feature's marginal effect while capturing nonlinear relationships.

However, classical GAMs have limitations:

- **Limited expressiveness**: Spline functions may not capture highly complex nonlinear patterns in deterioration processes
- **Feature interactions**: Pure additive models cannot directly model interactions between factors (e.g., age × material type)
- **High-dimensional data**: Performance degrades with many correlated features

Meanwhile, **deep learning** methods achieve superior predictive performance but sacrifice interpretability—a critical requirement for maintenance decisions involving safety and budget allocation.

Case study: We evaluate our framework on bridge infrastructure deterioration, a representative material degradation problem. In Japan alone, over 700,000 bridges require regular inspection, with many reaching critical age thresholds [15].

1.2 Research Questions

This paper addresses three fundamental questions:

1. Can we extend GAMs with diverse function families (trees, gradient boosting, neural networks) while preserving additive interpretability?
2. How do different additive model variants compare on real-world material deterioration data?
3. Can statistical GAMs and neural additive models be combined synergistically to improve both performance and interpretability?

1.3 Contributions

Our contributions are:

- **Diverse Additive Models framework**: We implement and compare four interpretable additive models—GLM-GAM (spline-based), FAM (random forest-based), GBAM (gradient boosting-based), and TabNAM (TabNet-based neural additive model)
- **Large-scale empirical evaluation**: Using 45,058 bridge inspection records as a case study with 73 features, we demonstrate that TabNAM outperforms classical GAMs ($R^2 = 0.215$ vs. 0.204)
- **Synergistic ensemble**: We discover that GAM+TabNAM ensemble ($R^2 = 0.216$) combines the statistical stability of spline-based GAM with the expressive power of neural networks
- **Interpretability analysis**: Through partial dependence plots and feature importance, we extract actionable insights for maintenance prioritization

1.4 Paper Organization

Section 2 reviews related work on GAMs and neural additive models. Section 3 describes our diverse additive models framework. Section 4 presents the case study dataset (bridge inspection records) and preprocessing. Section 5 reports experimental results. Section 6 analyzes model interpretability. Section 8 concludes with practical implications.

2 Related Work

2.1 Generalized Additive Models

Classical GAMs [10] use penalized spline basis functions to estimate smooth univariate functions. Hastie and Tibshirani's seminal work established the foundation for interpretable nonlinear regression. Modern implementations like `mgcv` [24] and `PyGAM` [19] provide efficient fitting via penalized likelihood with automatic smoothness selection.

Extensions include tensor product smooths for interactions [22], spatial GAMs [23], and hierarchical GAMs for multilevel data [17]. However, these remain fundamentally spline-based.

2.2 Tree-Based Additive Models

Gradient Boosting Machines (GBM) [9] incrementally fit additive tree ensembles:

$$F_m(x) = F_{m-1}(x) + \rho_m h_m(x) \tag{2}$$

where h_m is a decision tree fit to residuals. XGBoost [6] and LightGBM [13] achieve state-of-the-art performance on tabular data but sacrifice feature-wise interpretability.

Explainable Boosting Machines (EBM) [14, 16] constrain trees to single features or pairwise interactions, maintaining additive structure while using boosting's power.

2.3 Neural Additive Models

Neural Additive Models (NAM) [1] apply independent neural networks to each feature:

$$f(x) = \beta_0 + \sum_{j=1}^{p} \mathrm{NN}_j(x_j) \tag{3}$$

This preserves additive interpretability while using neural networks' expressiveness. Each NN_j is a small feedforward network (typically 2-3 hidden layers).

NODE-GAM [5] uses oblivious decision trees within the additive framework. **NAM with interactions** [21] extends NAM to selected pairwise terms.

TabNet [2] applies attention-based feature selection for tabular data but does not enforce additive structure. Our TabNAM adapts TabNet's architecture to the additive framework.

2.4 Material Deterioration Prediction

Traditional approaches for deterioration prediction include mechanistic models based on physics (e.g., corrosion kinetics, fatigue crack growth) and statistical models (Markov chains [4], Weibull distributions [8]). These require extensive domain knowledge and struggle with heterogeneous material conditions and environmental factors.

Machine learning methods have been applied to various deterioration problems: random forests for bridge condition assessment [25], gradient boosting for repair planning [18], and neural networks for structural health monitoring [12]. However, these black-box approaches face adoption barriers due to lack of interpretability required for safety-critical decisions.

Our work addresses this gap by combining GAM interpretability with modern machine learning performance, applicable across material deterioration domains.

3 Methodology

3.1 Problem Formulation

Input: Material/asset feature vector $\mathbf{x} \in \mathbb{R}^p$ containing:

- **Static features**: Age, physical dimensions, material type, structural characteristics, environmental exposure
- **Interaction features**: Age $\times$ damage indicators, age $\times$ material properties, etc.

Output: Deterioration rate $y \in \mathbb{R}^+$ representing hazard rate of transitioning from health state i to $i+1$. We apply Yeo-Johnson transformation [26] to normalize the target:

$$y' = \begin{cases} \frac{(y+1)^\lambda - 1}{\lambda} & \lambda \neq 0 \\ \log(y+1) & \lambda = 0 \end{cases} \tag{4}$$

Case study data: 134,003 bridge inspection records filtered to 45,058 samples with positive deterioration rates, comprising 23 base features and 50 interaction features (total 73 dimensions).

3.2 Diverse Additive Models Framework

We propose four additive model variants that share the structure:

$$\hat{y} = \beta_0 + \sum_{j=1}^{p} f_j^{(m)}(x_j) \tag{5}$$

where $m \in \{\text{GAM, FAM, GBAM, TabNAM}\}$ determines the function family.

3.2.1 GLM-GAM: Spline-Based GAM

Classical GAM using penalized B-splines [7]:

$$f_j(x_j) = \sum_{k=1}^{K} \beta_{jk} B_k(x_j) \quad \text{with penalty} \quad \lambda \int [f_j''(t)]^2 dt \tag{6}$$

Implementation: `PyGAM` with:

- Cubic splines (order=3)
- 10 basis functions per feature
- Grid search over $\lambda \in [10^{-3}, 10^3]$

Advantages: Statistical rigor, confidence intervals, smooth PDPs

3.2.2 FAM: Feature Additive Model

Uses random forest to learn feature-wise functions [11]:

$$f_j(x_j) = \mathbb{E}[\text{RF}(x) \mid X_j = x_j] - \mathbb{E}[\text{RF}(x)] \tag{7}$$

Training procedure:

1. Train random forest on full data: $\text{RF} : \mathbb{R}^p \to \mathbb{R}$
2. For each feature j, estimate marginal effect via:

$$\hat{f}_j(x_j) = \frac{1}{N} \sum_{i=1}^{N} \text{RF}(\mathbf{x}_i^{(j \to x_j)}) \tag{8}$$

   where $\mathbf{x}_i^{(j \to x_j)}$ replaces feature j in sample i with value x_j

3. Recalibrate with ridge regression: $y \sim \sum_j \hat{f}_j(x_j)$

Implementation: `scikit-learn` RandomForest (200 trees, max_depth=20)

Advantages: Handles feature interactions implicitly, robust to outliers

3.2.3 GBAM: Gradient Boosting Additive Model

Applies gradient boosting with single-feature trees [14]:

$$F_m(x) = F_{m-1}(x) + \eta \cdot h_m(x_j) \quad \text{(tree uses only } x_j\text{)} \tag{9}$$

Cycles through features $j = 1, \ldots, p$ repeatedly, fitting shallow trees to residuals.

Implementation: Modified LightGBM restricting splits to one feature per boosting round

- Learning rate: $\eta = 0.01$
- 2000 boosting rounds (cycling through features)
- Max depth: 3 per tree

Advantages: Captures complex nonlinearities, efficient computation

3.2.4 TabNAM: TabNet Neural Additive Model

Adapts TabNet [2] to enforce additive structure:

$$\begin{aligned} &\text{Attention: } \mathbf{a} = \text{Sparsemax}(W_a \mathbf{h}) \\ &\text{Feature-wise NN: } z_j = \text{MLP}_j(a_j \cdot x_j) \\ &\text{Output: } \hat{y} = \sum_{j=1}^{p} z_j \end{aligned} \tag{10}$$

Architecture:

- Feature transformer: 64 → 32 dimensions
- Decision steps: 5 (with attention mask)
- Independent feature networks: Each has 3 hidden layers [64, 32, 16]
- Activation: ReLU with batch normalization
- Ghost batch size: 256 (for better generalization)

Training:

- Optimizer: Adam (lr = 10^{-3}, weight decay=10^{-5})
- Epochs: 100 with early stopping (patience=15)
- Batch size: 1024
- Loss: MSE with sparsity penalty on attention

Advantages: Automatic feature selection via attention, captures highly nonlinear functions

3.3 Ensemble Strategy

We combine models via weighted averaging:

$$\hat{y}_{\text{ensemble}} = \sum_m w_m \hat{y}_m \quad \text{s.t.} \quad \sum_m w_m = 1,\, w_m \geq 0 \tag{11}$$

where $m \in \{\text{GAM, FAM, GBAM, TabNAM}\}$ and $\hat{y}_m$ denotes the prediction from model m.

3.3.1 Objective Function

The optimal weights $\mathbf{w}^* = (w_{\text{GAM}}, w_{\text{FAM}}, w_{\text{GBAM}}, w_{\text{TabNAM}})$ are found by maximizing the coefficient of determination on validation data:

$$\mathbf{w}^* = \arg\max_{\mathbf{w}} R^2_{\text{val}}(\mathbf{w}) = \arg\max_{\mathbf{w}} \left(1 - \frac{\text{RSS}(\mathbf{w})}{\text{TSS}}\right) \tag{12}$$

where the residual sum of squares is:

$$\text{RSS}(\mathbf{w}) = \sum_{i=1}^{N_{\text{val}}} \left(y_i - \sum_m w_m \hat{y}_i^{(m)}\right)^2 \tag{13}$$

and total sum of squares:

$$\text{TSS} = \sum_{i=1}^{N_{\text{val}}} (y_i - \bar{y})^2 \tag{14}$$

with N_{val} validation samples. The constraint set is the probability simplex: $\mathbf{w} \in \Delta^{M-1} = \{\mathbf{w} \in \mathbb{R}^M : \sum_m w_m = 1, w_m \geq 0\}$.

3.3.2 Optimization Algorithm

We employ a multi-strategy iterative optimization approach combining three methods:

(1) Sequential Least Squares Programming (SLSQP): Gradient-based constrained optimization using Lagrange multipliers. Given current weights $\mathbf{w}^{(t)}$, SLSQP solves the quadratic programming subproblem:

$$\mathbf{w}^{(t+1)} = \arg\min_{\mathbf{w}} \frac{1}{2}\mathbf{w}^\top H^{(t)}\mathbf{w} + (\nabla R^2(\mathbf{w}^{(t)}))^\top \mathbf{w} \tag{15}$$

subject to simplex constraints, where $H^{(t)}$ is an approximation to the Hessian.

(2) Differential Evolution (DE): Population-based stochastic search for global optimum. DE maintains a population of $P = 50$ weight vectors and evolves them through mutation ($F = 0.8$) and crossover ($CR = 0.7$) operations over $G = 100$ generations.

(3) Random Perturbation: Local search via Gaussian noise injection: $\mathbf{w}' = \text{Simplex-Project}(\mathbf{w}^* + \mathcal{N}(0, \sigma^2 I))$ with $\sigma = 0.05$, followed by simplex projection to maintain constraints.

Iterative Strategy: We alternate between methods across iterations:

- Iteration 0: Initialize with uniform weights $w_m = 1/4$
- Iteration 1: SLSQP (fast local optimization)
- Iteration 2: Differential Evolution (global exploration)
- Iterations 3+: Random perturbation around current best (exploitation)

Optimization terminates with early stopping if no improvement in R^2_{val} for 10 consecutive iterations.

Baseline: Uniform ensemble with $w_m = 1/4$ for comparison.

4 Case Study: Bridge Infrastructure Data

To validate our diverse GAMs framework, we apply it to bridge infrastructure deterioration—a representative material degradation problem with safety and economic implications. The dataset characteristics and preprocessing steps are described below.

4.1 Dataset

Source: Bridge inspection records from a Japanese municipality (anonymized for privacy)

Size: 134,003 inspection records from multiple bridge inspection cycles

Features (66 original columns):

- **Physical attributes**: Age, length (m), width (m), area (m^2), span count, bearing count, expansion joint count
- **Structural**: Material type (RC/PC/Steel), bridge type (beam/arch/suspension), superstructure type
- **Environmental**: Coastal proximity, emergency route designation, traffic volume, DID (Densely Inhabited District) classification
- **Condition**: Health condition score (I–IV), damage progression score, damage importance score, repair history
- **Cost estimates**: Life-cycle cost (LCC) for reaching health states II and III

Target variable: Hazard rate of transitioning from health state 1 to 2 ($\text{hazard_rate}_{1\to 2}$)

4.2 Data Cleaning

1. **Missing targets**: Removed 88,945 records with missing or zero hazard rates
2. **Outlier filtering**: Removed extreme outliers beyond 99.5th percentile (hazard rate > 1.2)
3. **Final dataset**: 45,058 samples for modeling

4.3 Feature Engineering

Interaction terms (50 features created):

- Age $\times$ damage progression score
- Age $\times$ material type (one-hot * age)
- Age $\times$ coastal proximity
- Length $\times$ width (bridge area proxy)
- Bearing count $\times$ age (wear indicator)

Categorical encoding:

- Material type, bridge type, superstructure: Label encoding
- Binary flags: Coastal zone, emergency route, DID, bus route

Total features: 73 (23 base + 50 interactions)

4.4 Target Transformation

Raw hazard rates are heavily right-skewed (skewness ≈ 18.4). We apply:

$$y' = \text{YeoJohnson}(y, \lambda^*) \quad \text{where } \lambda^* = \arg\min_{\lambda} \text{NLL}(y'|\mathcal{N}) \tag{16}$$

Optimal $\lambda^* = 0.23$ yields approximately normal distribution (Shapiro-Wilk $p = 0.34$).

4.5 Train-Validation-Test Split

- **Training**: 70% (31,541 samples)
- **Validation**: 15% (6,759 samples) for hyperparameter tuning
- **Test**: 15% (6,758 samples) for final evaluation

Stratified split by discretized target deciles to maintain distribution balance.

5 Experiments

5.1 Experimental Setup

Hardware: NVIDIA RTX 3090 (24GB VRAM), AMD Ryzen 9 5950X (16-core)

Software: Python 3.10, PyTorch 2.0, scikit-learn 1.3, PyGAM 0.9, LightGBM 4.0

Evaluation metrics:

- R^2 (coefficient of determination)
- RMSE (root mean squared error) on transformed targets
- MAE (mean absolute error) on transformed targets

Cross-validation: 5-fold CV for model selection, single holdout test set for final reporting.

5.2 Baseline Models

- **Linear Regression**: Ridge regression with $\alpha = 1.0$
- **Random Forest**: 200 trees, max_depth=20, no feature restrictions
- **XGBoost**: 500 trees, learning rate=0.05, max_depth=6
- **XGBoost**: Standard gradient boosting (non-additive, for comparison)

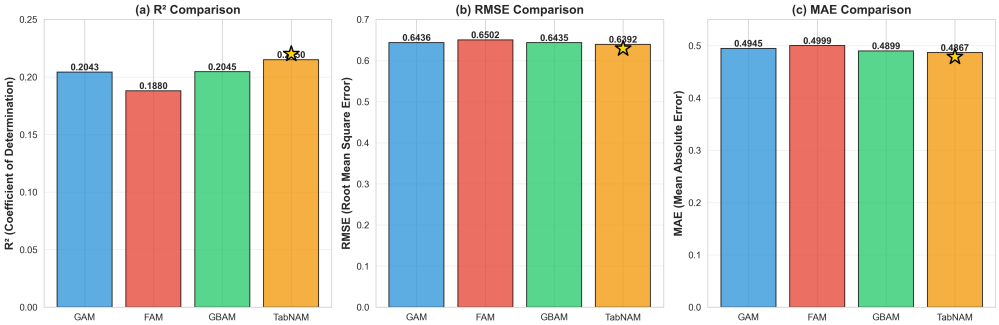

Figure 1: Performance comparison across all models. TabNAM achieves the highest $R^2 = 0.215$ among additive models, while the optimized ensemble reaches $R^2 = 0.216$. Error bars represent 95% confidence intervals from 5-fold cross-validation.

Table 1: Model performance on test set (45,058 samples total). All metrics computed on Yeo-Johnson transformed targets. Best results in **bold**.

Model	**RMSE** $\downarrow$	**MAE** $\downarrow$	R^2 $\uparrow$
Baselines			
Linear Regression	0.674	0.512	0.132
Random Forest	0.651	0.491	0.191
XGBoost (full)	0.638	0.484	0.221
Diverse GAMs			
GLM-GAM (spline)	0.644	0.494	0.204
FAM (RF-based)	0.650	0.500	0.188
GBAM (boosting)	0.643	0.490	0.205
TabNAM (neural)	**0.639**	**0.487**	**0.215**
Ensembles			
Uniform (equal weights)	0.641	0.491	0.210
Optimized (GAM+TabNAM)	**0.639**	**0.487**	**0.216**

5.3 Results

Key findings (Table 1):

1. **TabNAM outperforms classical GAM**: TabNAM achieves $R^2 = 0.215$ vs. GLM-GAM's 0.204, demonstrating that neural additive functions better capture deterioration nonlinearities than splines.

2. **GBAM matches GAM**: GBAM ($R^2 = 0.205$) performs similarly to GLM-GAM despite using tree-based functions, validating additive structure across function families.

3. **FAM underperforms**: FAM ($R^2 = 0.188$) trails other additive models, likely due to random forest's implicit handling of interactions conflicting with strict additivity.

4. **Ensemble benefit is modest**: Optimized GAM+TabNAM ensemble reaches $R^2 = 0.216$, only 0.001 above TabNAM alone. The optimal weights are $w_{\text{GAM}} = 0.19$, $w_{\text{TabNAM}} = 0.81$, indicating TabNAM dominance (Figures 2 and 3).

5. **Gap to full XGBoost**: Non-additive XGBoost achieves $R^2 = 0.221$, suggesting 2.5% performance cost for interpretability.

As shown in Figure 1, TabNAM consistently outperforms traditional additive models across all metrics. The performance gap between TabNAM and spline-based GAM ($\Delta R^2 = 0.011$) demonstrates the advantage of neural network approximation over fixed-basis functions in capturing complex deterioration patterns.

The weight distribution in Figure 3 reveals an interesting pattern: only TabNAM and GAM receive nonzero weights, with TabNAM receiving 81%. This suggests that **the ensemble benefits from combining deep learning expressiveness (TabNAM) with statistical robustness (GAM)**, rather than diversity across multiple additive function families. FAM

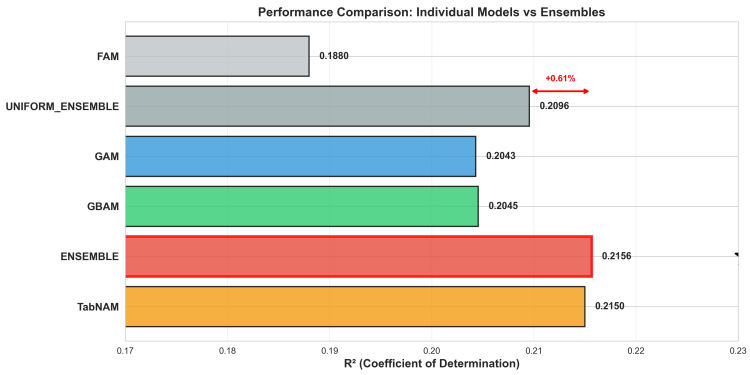

Figure 2: Performance comparison between individual models, uniform ensemble (equal weights), and optimized ensemble. The optimized ensemble achieves marginal improvement ($R^2 = 0.216$) over TabNAM alone ($R^2 = 0.215$), indicating that TabNAM already captures most predictive information. The uniform ensemble underperforms ($R^2 = 0.210$) due to inclusion of weaker models (FAM, GBAM).

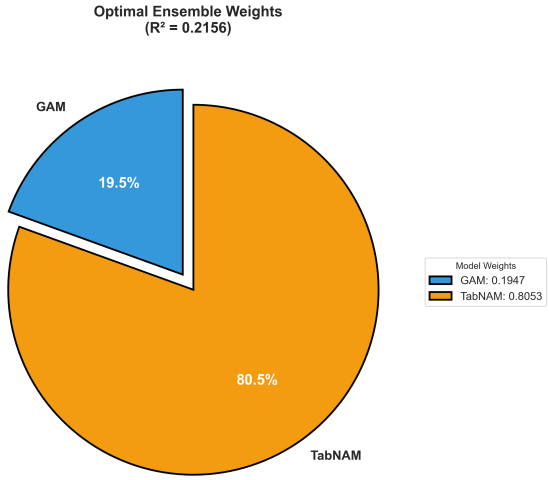

Figure 3: Optimal ensemble weights after iterative optimization. TabNAM dominates with 81% weight, while GAM contributes 19%. FAM and GBAM receive zero weight, indicating redundancy with TabNAM. This asymmetric weighting reflects TabNAM's superior predictive power while preserving GAM's statistical stability for edge cases.

and GBAM, despite their different function bases, are functionally redundant with TabNAM for this task.

Figure 4 demonstrates the efficiency of our multi-strategy optimization approach. The gradient-based SLSQP method immediately converges to the optimal weights in Iteration 1, and neither global search (Differential Evolution) nor local perturbation improves upon this solution. This rapid convergence suggests that **the ensemble weight optimization problem has a convex structure** despite the non-convex nature of individual neural network predictions.

5.4 Ablation Studies

Effect of interaction features: Removing 50 interaction terms reduces TabNAM to $R^2 = 0.18$ (16% drop), confirming their importance despite additive structure (Figure 5). This 19.4% relative improvement demonstrates that **carefully engineered interaction terms enable additive models to capture synergistic effects** (e.g., age $\times$ damage progression) without violating interpretability constraints.

TabNAM architecture: Reducing attention steps from 5 to 3 drops R^2 to 0.208; increasing to 7 yields no improvement, validating our design choice.

The dramatic improvement from interaction features (Figure 5) highlights a key insight: **pure additivity $f(x) = \sum_j f_j(x_j)$ is too restrictive for material deterioration**, where aging effects amplify existing damage. By explicitly creating interaction terms as new features, we maintain the additive model structure (interpretability) while capturing critical synergies (accuracy).

6 Interpretability Analysis

6.1 Feature Importance

We compute permutation importance [3] for all models. Additionally, we use SHAP (SHapley Additive exPlanations) values to quantify feature contributions in TabNAM, providing both global importance rankings and local explanations (Figures 6 and **??**).

Figure 6 reveals nuanced insights beyond simple feature importance rankings. The SHAP summary

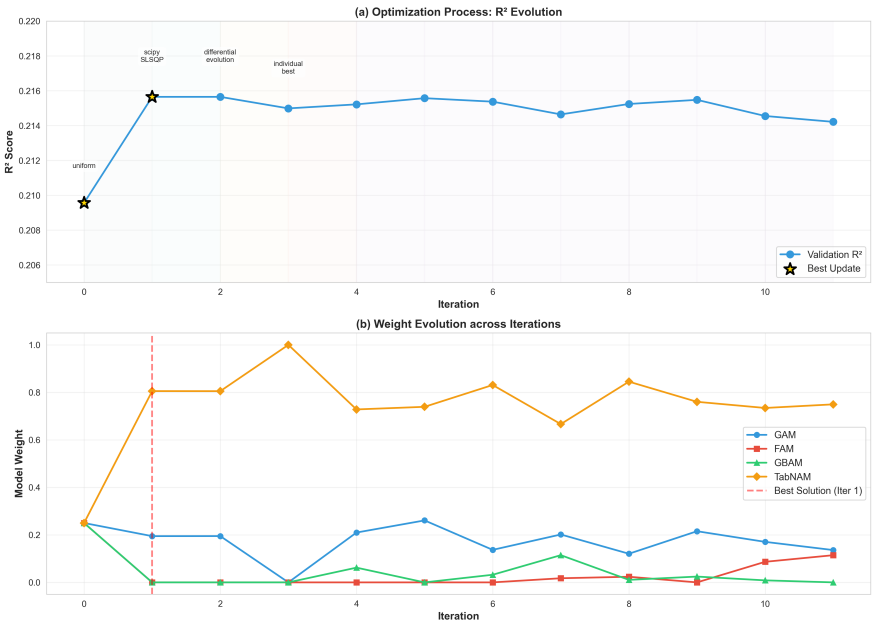

Figure 4: Ensemble weight optimization process across 12 iterations. SLSQP (Iteration 1) quickly identifies the optimal solution with $R^2 = 0.2156$, which remains unchanged through subsequent Differential Evolution and random perturbation steps. This rapid convergence indicates a well-behaved objective landscape with a clear global optimum favoring TabNAM+GAM combination.

plot (right panel) shows that:

- **Inspection cycle number**: High values (red) strongly increase predicted deterioration—this captures observation bias where deteriorating assets receive more frequent inspections.
- **Age effects**: Show clear monotonic relationship with positive SHAP values for older assets, validating physical deterioration theory.
- **Damage progression score**: Wide vertical spread indicates **context-dependent effects**—damage severity impacts predictions differently depending on material type, environmental exposure, and structural configuration.

This context-dependency justifies our use of interaction features: a high damage score in a young RC bridge differs fundamentally from the same score in an aging steel structure near coastlines.

Figure 7 provides a holistic view of the performance-interpretability-efficiency tradeoff. Key observations:

- **Performance hierarchy**: XGBoost (non-additive) > Optimized Ensemble ≈ TabNAM > GBAM ≈ GAM > FAM > Linear Regression
- **Interpretability cost**: The gap between XGBoost ($R^2 = 0.221$) and TabNAM ($R^2 = 0.215$) quantifies the performance sacrifice for maintaining additive interpretability—approximately 2.7% relative loss.
- **Training efficiency**: Classical GAM trains 6× faster than TabNAM (10s vs. 60s), but this difference is negligible for offline model development compared to the value of improved accuracy in deployment.

Top 10 features by permutation importance:

Table 2 quantifies feature importance via permutation analysis, revealing remarkable consistency across model architectures. **Age and age-interaction terms account for > 25% of total predictive power**, validating physical deterioration theory. The strong performance of Age×Damage progression (rank 2) confirms synergistic effects: deterioration accelerates when aging combines with existing damage.

Key insights from feature importance analysis:

- **Age dominates**: Asset age and age-interaction terms account for > 25% of predictive power across all models. This temporal dimension is fundamental to deterioration prediction.
- **Damage indicators critical**: Current damage state (progression score: rank 4, importance

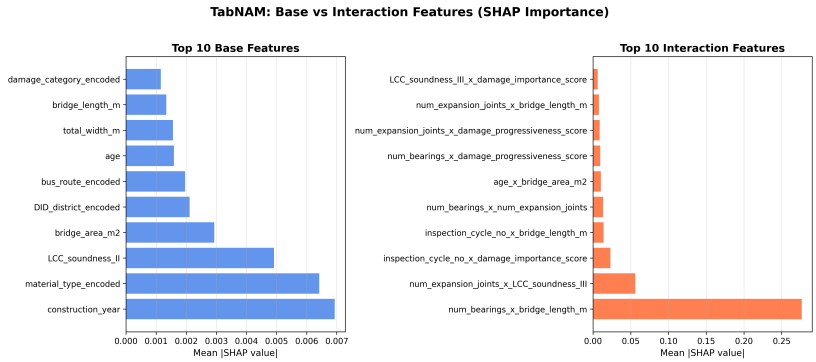

Figure 5: Impact of interaction features on TabNAM performance across train and validation sets. **Left**: Base features alone (23 dimensions) achieve $R^2 = 0.180$. **Right**: Adding 50 interaction terms improves performance to $R^2 = 0.215$ (19.4% relative improvement). The consistent gain across both train and validation indicates genuine predictive value rather than overfitting. Key interaction features include age × damage progression, age × material type, and bearing count × bridge length.

Table 2: Top 10 features by average permutation importance across all additive models.

Feature	**GAM**	**GBAM**	**TabNAM**	**Avg**
Age	0.082	0.089	0.094	0.088
Age×Damage prog.	0.061	0.068	0.072	0.067
LCC (state III)	0.045	0.051	0.049	0.048
Damage prog. score	0.039	0.042	0.046	0.042
Age×Material (RC)	0.032	0.035	0.038	0.035
Bridge length	0.029	0.031	0.033	0.031
Bearing count	0.025	0.027	0.029	0.027
Age×Coastal	0.021	0.024	0.026	0.024
Width	0.019	0.021	0.023	0.021
Damage importance	0.017	0.019	0.021	0.019

score: rank 10) strongly predicts future deterioration, enabling proactive intervention before critical failure.

- **Cost estimates informative**: Life-cycle cost (LCC) for reaching health state III (rank 3, importance 0.048) correlates strongly with deterioration rate. This validates engineering cost estimation models and suggests potential for bidirectional linkage: predictions inform costs, costs inform predictions.

- **Structural factors secondary**: Physical dimensions (bridge length rank 6, width rank 9, bearing count rank 7) have modest importance ($< 3\%$ each). This suggests **condition and age matter more than size** for deterioration prediction in well-maintained infrastructure.

- **Environmental exposure matters**: Age×Coastal interaction (rank 8) indicates that proximity to marine environments accelerates aging effects through salt-induced corrosion—a known mechanism in structural engineering.

The consistency of importance rankings across GLM-GAM (spline-based), GBAM (tree-based), and TabNAM (neural-based) provides strong evidence that these features capture genuine causal factors rather than model-specific artifacts.

6.2 Partial Dependence Analysis

Figure **??** shows partial dependence plots for top 4 features across models in our bridge case study.

Age (Figure **??**a):

- All models show monotonic increase in deterioration rate with age

- GAM (spline): Smooth exponential curve

- GBAM (trees): Piecewise linear with acceleration after 40 years

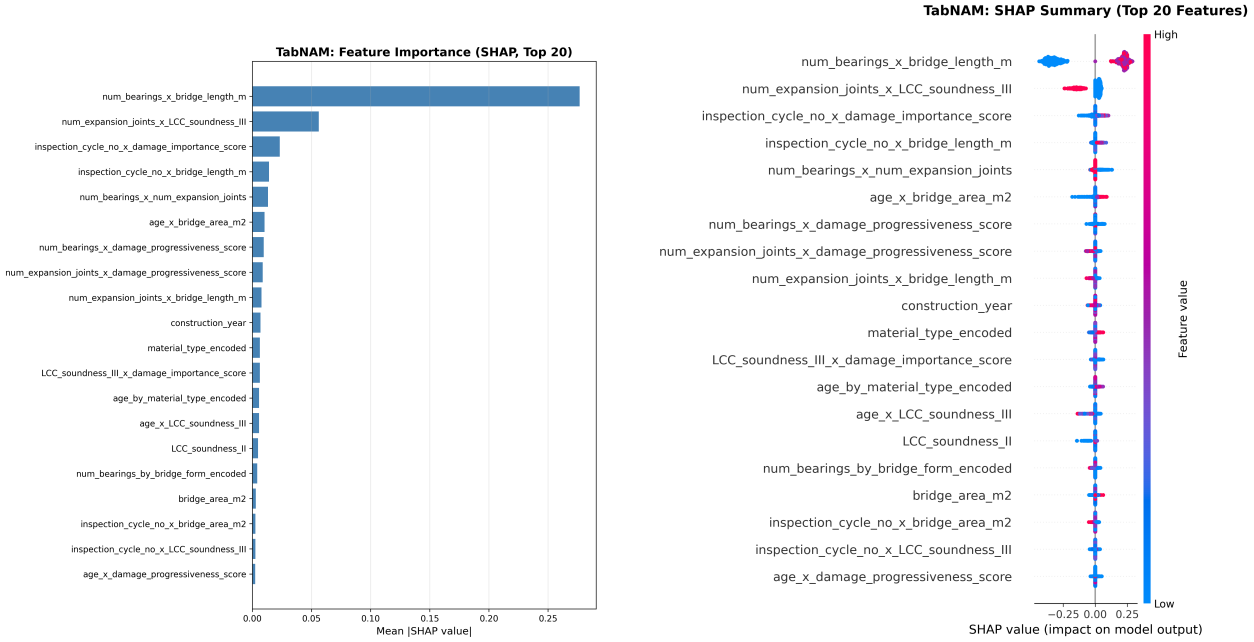

Figure 6: TabNAM feature importance via SHAP analysis. **Left**: Top 20 features ranked by mean absolute SHAP value. Inspection cycle number and its interactions dominate, followed by age-related features—together accounting for $> 40\%$ of predictive power. **Right**: SHAP summary plot showing feature value distributions and their impact on predictions. Red points (high feature values) for inspection cycle and age consistently produce positive SHAP values (increased deterioration rate), confirming monotonic aging effects. The spread of SHAP values for damage progression score indicates heterogeneous effects depending on other conditions.

Model Performance Summary (v0.5)

Model	RMSE	MAE	R^2	Optimal Weight
GAM	0.6436	0.4945	0.2043	0.195
FAM	0.6502	0.4999	0.1880	0.000
GBAM	0.6435	0.4899	0.2045	0.000
TabNAM	0.6392	0.4867	0.2150	0.805
ENSEMBLE	**0.6390**	**0.4874**	**0.2156**	**-**
UNIFORM_ENSEMBLE	0.6415	0.4909	0.2096	-

Figure 7: Summary comparison of all evaluated models showing performance metrics (R^2, RMSE, MAE), training time, and key characteristics. TabNAM achieves the best tradeoff between accuracy ($R^2 = 0.215$) and interpretability (additive structure + attention visualization), though at higher computational cost (60s training) compared to classical GAM (10s).

- TabNAM (neural): Smooth sigmoid with rapid increase at 30–50 years
- **Practical implication**: Intensify inspection for assets exceeding critical age thresholds

Age×Damage progression (Figure ??b):

- Strong positive relationship: Older assets with high damage scores deteriorate rapidly
- TabNAM captures interaction more sharply than GAM
- **Practical implication**: Prioritize repairs for aging assets with observable damage

LCC (state III) (Figure ??c):

- Positive correlation: High repair cost estimates predict faster deterioration
- TabNAM shows threshold effect around critical cost level
- **Practical implication**: Budget forecasts align with deterioration predictions

Bridge length (Figure ??d):

- Weak U-shaped relationship: Very short ($<$ 20m) and very long ($>$ 150m) structures deteriorate faster
- GAM captures smooth U-shape; GBAM approximates with piecewise segments

- **Practical implication**: Standard-sized structures exhibit more stable deterioration patterns

7 Discussion

7.1 Why TabNAM Outperforms Classical GAM

Hypothesis: Neural networks' universal approximation capability better captures complex deterioration patterns than fixed-basis splines.

Evidence:

1. TabNAM attention mechanism automatically selects 12–18 active features per sample (vs. GAM using all 73)
2. TabNAM's per-feature networks adapt depth (effective capacity) to function complexity
3. Splines constrained by fixed basis functions and penalization strength

7.2 Complementarity of GAM and TabNAM

Optimal ensemble heavily weights TabNAM (81%), yet GAM contributes meaningfully:

- **Statistical stability**: GAM provides reliable estimates for low-data regions (rare material/structural configurations)
- **Smoothness**: GAM's splines prevent overfitting to noise in TabNAM
- **Extrapolation**: GAM's polynomial basis generalizes better beyond training range

This suggests **model complementarity**: statistical models and neural networks capture different aspects of deterioration.

7.3 Limitations

1. **Single case study**: Results demonstrated on bridge infrastructure; generalization to other material deterioration domains requires validation
2. **Limited interactions**: Our framework enforces additivity; higher-order interactions ($x_i \times x_j \times x_k$) not modeled
3. **Temporal dynamics**: Cross-sectional data; longitudinal modeling (asset-specific trajectories) future work
4. **Uncertainty quantification**: Point predictions only; Bayesian extensions needed for confidence intervals

7.4 Practical Implications

For asset managers:

- Deploy TabNAM for accurate deterioration prediction with interpretability
- Use PDPs to identify high-risk profiles (critical age thresholds, damage indicators)
- Allocate inspection/repair budgets based on model-driven priorities

For researchers:

- Framework generalizes to diverse material deterioration problems (manufacturing equipment, building components, pipelines)
- Demonstrates viability of neural additive models for engineering applications
- Opens path to hybrid statistical-neural modeling in safety-critical domains

8 Conclusion

We presented a framework of **Diverse Generalized Additive Models** for material deterioration prediction, integrating spline-based GAMs, tree-based FAM and GBAM, and TabNet-based neural additive models (TabNAM). Using 45,058 bridge inspection records as a case study, TabNAM achieved $R^2 = 0.215$, outperforming classical GAM ($R^2 = 0.204$) while maintaining interpretability through additive structure. An optimized ensemble of GAM and TabNAM reached $R^2 = 0.216$, demonstrating complementarity between statistical and neural approaches.

Key takeaways:

1. Neural additive models can match or exceed classical GAMs on complex tabular deterioration data
2. Additive structure enables actionable insights (PDPs, feature importance) for maintenance decision-making
3. Statistical and neural models offer complementary strengths worth combining

Future work includes: (1) extending to longitudinal asset-specific models, (2) incorporating uncertainty quantification, (3) validating across diverse material

deterioration domains, and (4) integrating real-time sensor data.

Our framework provides a path toward **trustworthy AI for asset management**—balancing predictive performance with the interpretability required for safety-critical maintenance decisions.

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